Kananelo Ketlele

Test lead · GitHub: Kananelok · Student no. 202401765

Responsible for the automated tests that show the assembler is correct.

Files and document sections

File / section	Content
`tests/run.js`	The test script, run with node tests/run.js; 37 checks, all passing.
`docs/M2-design.md §9`	Description of what the tests cover.
`tests (planned, M3 and M4)`	Tests for the control decoder, pipeline engine, forwarding and hazard unit.

Work allocated

- Test harness with a check() helper, hexOf() for single-line encodings and expectError() that verifies both the error text and the line number.
- Group 1: the six M1 worked examples reproduced exactly (e.g. 0x00C48040, 0x0150C080, 0x050000C0, 0x10C40000, 0x14480080, 0x18480030).
- Group 2: every instruction has the correct opcode or funct value.
- Group 3: negative, hexadecimal and limit immediates, and forward and backward branch displacements.
- Group 4: labels, comments, blank lines and a full array-sum program.
- Group 5: eight error cases with plain-language messages: unknown instruction, bad register, wrong operand count, immediate out of range, undefined label, duplicate label, bad memory syntax and curly apostrophe.
- Test plan for M3 and M4: control-signal table for all 15 instructions and the expected stall and flush counts of the hazard examples.